

Ashutosh Naik

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EDUCATION

University of Colorado Boulder

Master of Science in Computer Science

Relevant Coursework: Datacentre Scale Computing, Design and Analysis of Algorithms

Boulder, CO

Expected May 2024

Sardar Patel Institute of Technology

Bachelor of Technology in Computer Engineering GPA: 9.77/10

Relevant Coursework: Object Oriented Programming, Artificial Intelligence and Soft Computing, Distributed Systems, Design and Analysis of Algorithms, Database Management Systems, Advanced Data Structures

Mumbai, India

May 2022

TECHNICAL SKILLS

Programming Languages: Python | C++ | C | SQL

Libraries and Frameworks: Docker | Kubernetes | Google Cloud Platform | MPI | pandas | matplotlib | TensorFlow | PyTorch | NumPy

EXPERIENCE

ML Intern, Skinzy Software Solutions

December 2020 – December 2021

- Developed an object detection and segmentation model to better identify and distinguish across more than 15 skin diseases.
- Fine-tuned neural network models such as MRCNN, UNET and utilized detectron2 library on custom datasets for segmenting skin diseases. Acquired an accuracy of over 90% across each class.
- Attained an accuracy of over 80% on jaundice detection task using PyTorch.
- Conducted a session and was in-charge of hiring and training new interns. Also collaborated with the App Dev team to identify major payment related bug in application.

PROJECTS

Image Augmentation as a Service

November 2022

- Designed and developed an application as a service that takes an image and applies various transformation functions on it.
- Implemented the individual augmentation functions as FaaS using Google Cloud Functions. Images are stored and retrieved from Google Cloud Storage.
- Utilized Flask to create Rest frontend which was deployed in a container running on a Kubernetes cluster.

Segmentation of Nuclei using Transformer based Architecture

January 2022 – April 2022

- Pre-processed 4 publicly available nuclei segmentation datasets obtained under different laboratory conditions using OpenCV, NumPy and scikit-image.
- Fine-tuned the TransUnet model using PyTorch on the datasets obtained.
- Obtained a 64% higher IOU score than U-Net - a CNN based segmentation model, under similar training conditions.

Vigila – Application for General Safety

January 2021 - April 2021

- Designed and developed prototype for a community driven safety application in Flutter and Firebase.
- Conducted a survey of over 375 people to evaluate public opinion of such application.
- Accepted at IEEE 2021 International Conference on Advances in Computing, Communication and Control (ICAC3'21) 7th Edition conference.

Neuro-Evolution Snake

June 2020

- Implemented classic snake game in python using pygame library.
- Implemented a genetic algorithm - a neural network using concept of neuro-evolution to 'see' in 8 directions using NumPy.
- After a few generations network acquired ability to play, steering to avoid snake from hitting itself or surrounding walls and was able to score points by eating food.

Sudoku Solver

May 2020

- Employed OpenCV library to extract sudoku grid from page and identify cells containing digits.
- Trained a neural network using TensorFlow to identify extracted numbers from image.
- Utilized backtracking algorithm to solve actual sudoku grid.

CERTIFICATIONS

- "Natural Language Processing in TensorFlow", Coursera, June 2020
- "Convolutional Neural Networks in TensorFlow", Coursera, May 2020
- "Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning", Coursera, May 2020
- "Deep Learning Specialization", Coursera, May 2020